

Ashutosh Sudhir Kumbhar

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EDUCATION

Master of Software Engineering

Arizona State University

August 2024 - May 2026

Tempe, AZ, USA

Interests: Full Stack Software Development, Systems Engineering, Data Processing at Scale, Site Reliability, High Performance Computing

EXPERIENCE

Software Developer Intern – Synopsys

May 2025 – May 2026

- Architected scalable HPC simulation pipelines for semiconductor design workflows on AWS EKS using Terraform and Kubernetes, integrating RabbitMQ and KEDA for event-driven job scheduling across 15 Linux nodes processing 30K+ daily jobs for clients including Apple, Nvidia, MicroChip and many leading semiconductor companies.
- Debugged, monitored, and optimized production Kubernetes clusters using Datadog observability, reducing queue latency and deployment time while improving job throughput.
- Built and documented a Nest.js (TypeScript) API service for on-demand HPC provisioning, managing Terraform state via the Spacelift GraphQL API to execute AWS infrastructure operations at scale.
- Hardened production clusters with Kyverno policy enforcement, remediating security vulnerabilities and enforcing best practices.

Software Developer - Western Union Financial Services

January 2023 – August 2024

- Spearheaded an internal POS utility tool enabling proxy connections between environments, boosting developer productivity by 30%
- Engineered the WUPOS Database-Compare tool to automate configuration comparison between Production and QA/UAT, enabling developers to debug and replicate production issues 25% faster
- Integrated a Generative AI (LLM) summarization layer presenting configuration drift with probable root causes, accelerating root-cause analysis across 5+ SCRUM teams
- Resolved critical production bugs across services handling millions of financial transactions, improving system stability by 15%
- Migrated a legacy monolithic application to microservices using Angular 18 and Spring Boot; applied TDD with Playwright, raising test coverage by 40%

Full-Stack Software Developer - MindsClik Digital Solutions

January 2022 - January 2023

- Architected a white-label eCommerce platform with Flutter, building a shared component library (product pages, cart, checkout, payments) that allowed new client stores to be assembled from pre-built, pre-tested modules, reducing launch time by 60%
- Engineered intuitive, reusable UI components in Flutter, enhancing user experience and achieving a 4.8/5 satisfaction rating in usability.
- Enhanced data retrieval and API interactions with GraphQL using PostgreSQL, achieving a 60% improvement in response times.
- Automated provisioning of dedicated cloud environments for each eCommerce customer on AWS using Terraform Infrastructure as Code modules, configuring tenant-specific EC2 instances, S3 buckets, and VPC networking.

PROJECT EXPERIENCE

Code-Studio - Kubernetes-Based Lab-as-a-Service and Online Compiler: <https://hub.docker.com/r/8766834856/compiler>

- Designed and deployed a Kubernetes-based compute platform on Linux, providing isolated, on-demand dev environments for 200+ students.
- Developed a browser-based IDE using Vue.js, Node.js, and node-pty, supporting interactive Linux terminals, 10+ programming languages, and seamless integration with learning management systems such as Moodle and Canvas.
- Automated the provisioning, monitoring, and cleanup of containerized coding sessions while enforcing CPU, memory, and execution-time limits for workload isolation.
- Diagnosed Linux, container networking, and Kubernetes scheduling failures using logs and pod events, improving system reliability.

Restaurant Point of Sale Application (POS): <https://tasteandtwist.co.uk/>

- Engineered Angular NX powered Monorepo POS app, streamlined restaurant workflows via Firebase Cloud Functions and Firestore.
- Conceptualized and implemented a loyalty program system with integrated Email/SMS marketing capabilities, boosting customer retention.
- Devised a client-facing kiosk application mirroring McDonald's self-service functionality, simplifying staff workload by 30%.

Wolverine - Diagnostic Center Management System: pawnaaidiagnosticcenterpatan.netlify.app

- Engineered a patient referral and resource management system, promoting operational productivity by 30% through automated workflows.
- Developed a cross-platform app operating Vue.js and Electron, delivering a seamless PWA experience for users with 99% uptime reliability.

SKILLS

Technologies: C, C++, Java, Python, JavaScript, SQL, HTML, Algorithms, Data Structures, Object-Oriented Programming, Design Patterns, OS

Backend & Testing: Django, Spring MVC, Playwright, Node.js, GraphQL, Junit, A/B Testing, E2E Testing, Selenium

Frontend & Mobile Frameworks: Angular, ReactJS, Vue.js, Next.js, Electron.js, Flutter, React Native

Cloud & DevOps: Git, AWS, Docker, Kubernetes, Jenkins, Linux, RabbitMQ, Redis, Apache Kafka, Lambda, Terraform, Datadog, Prometheus

Database & Operations: MongoDB, PostgreSQL, Oracle, MySQL, ETL Framework, Job Schedulers, Jira, Agile, Project Management

ML and Artificial intelligence: LLM, MCP, Agentic AI, Retrieval Augmented Generation, Vector Databases, Knowledge Graphs

AWARDS & HONORS

Merit-Based scholarship from Arizona State University worth \$12000.